

Software Engineering: Requirements Engineering & UML Modelling
Problem Statement #27 | Campus & Academic Operations: Smart Cities, Transport & Logistics Warehouse Inventory & Pallet Location Tracker

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**PROBLEM STATEMENT #27 | Smart Cities,
Transport & Logistics Warehouse Inventory
& Pallet Location Tracker**

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1.Functional Requirements (FR-001 to FR-005)

Req ID	Description	Priority	Acceptance Criteria	Rationale
FR-001	The system shall validate proposed pallet placement coordinates against the rack's maximum load capacity before confirming storage.	High	Pass: Pallet is assigned to a valid rack where the total weight remains within the rated structural limit. Fail: System prevents placement when the total weight would exceed the rack's maximum capacity.	Prevents rack overloading and ensures safe and structurally compliant pallet storage.
FR-002	The system shall allow the Warehouse Operator to record and maintain the 3D location of each pallet using aisle, rack, level, and slot coordinates.	High	Pass: Pallet is stored with valid aisle, rack, level, and slot coordinates. Fail: System rejects storage when the location is invalid, unavailable, or incomplete.	Ensures accurate pallet positioning and enables efficient inventory retrieval.
FR-003	The system shall identify pallets and record stock movements using barcode or RFID scanning for inbound, outbound, and transfer operations.	High	Pass: Valid barcode/RFID scan identifies the pallet and records the corresponding stock movement. Fail: System rejects invalid or duplicate scans and displays an appropriate error message.	Automates inventory data capture, reduces manual entry errors, and improves movement accuracy.
FR-004	The system shall maintain a time-stamped history of stock movements including pallet ID, movement type, Warehouse Operator ID, and source and destination locations.	Medium	Pass: Each completed movement is stored with all required details and a timestamp. Fail: System does not complete the movement record when critical information is missing.	Provides traceability and an audit trail for all pallet movements within the warehouse.
FR-005	The system shall allow authorized users to search for an SKU or pallet ID and display its current 3D warehouse location and inventory status.	Medium	Pass: A valid SKU or pallet ID returns the correct current location and inventory status. Fail: System displays an appropriate message when no matching pallet is found.	Enables warehouse personnel to quickly locate inventory and improves operational efficiency.

2. Non-Functional Requirements (NFR-001 & NFR-002)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Scalability	The warehouse item lookup service shall locate and return the current SKU pallet coordinate across 100,000 bins within 100 ms under simulated peak-load conditions.	High	Pass: Benchmarking confirms that SKU/pallet location queries return results within 100 ms across 100,000 bins under peak-load testing. Fail: Query response time exceeds 100 ms during the required benchmark.	Ensures warehouse operators and supervisors can locate inventory quickly even when the system manages a large number of storage bins.
NFR-002	Security & Access Control	The system shall enforce Role-Based Access Control (RBAC) for Warehouse Operators and Logistics Supervisors and encrypt sensitive inventory and user data both at rest and in transit.	High	Pass: Users can access only functions permitted by their assigned role, and encrypted communication is verified during security testing. Fail: An unauthorized user can access restricted inventory functions or sensitive data.	Protects inventory information, prevents unauthorized modifications, and ensures secure access to warehouse operations.

3. Core Use-Case Flow Specification

Field	Specification
Use Case Name & ID	UC-01: Record Pallet Storage
Goal	To enable a Warehouse Operator to scan a pallet, assign it to a valid 3D shelf location, verify that the rack can safely support the pallet's weight, and record the storage operation for future tracking and retrieval.
Primary Actor	Warehouse Operator
Secondary Actors	Logistics Supervisor, Barcode/RFID Scanner
Preconditions	1. Warehouse Operator is authenticated and logged into the warehouse management system.2. The pallet has a valid barcode or RFID tag registered in the system.3. The proposed rack, level, and slot are available for pallet placement.4. Rack maximum weight capacity is available in the system.
Postconditions	1. The pallet is assigned a valid 3D warehouse coordinate (aisle, rack, level, slot).2. The system records the pallet's storage location and weight.3. A time-stamped storage movement record is created.4. The pallet becomes searchable using its SKU or pallet ID.
Main Success Scenario (MSS)	1. Warehouse Operator selects ' Store Pallet '.2. Operator scans the pallet's barcode/RFID tag (UC-1.1 «include»).3. System retrieves the pallet ID, SKU, and weight from the inventory database.4. Operator enters or selects the proposed aisle, rack, level, and slot coordinates.5. System validates that the selected 3D location is available and checks the rack's remaining load capacity (UC-1.2 «include»).6. System confirms that the pallet weight does not exceed the rack's maximum structural limit.7. System records the pallet location and creates a time-stamped stock movement record.8. System displays ' Pallet Stored Successfully ' along with the pallet ID and assigned 3D coordinates.
Alternate Flows	5a. Rack Weight Capacity Exceeded: 1. System calculates that adding the pallet would exceed the rack's maximum load capacity.2. System rejects the proposed placement and displays an overload warning.3. Operator selects another valid rack/location.4. Flow resumes at Step 4 of the Main Success Scenario. 4a. Location Unavailable: 1. System detects that the selected slot is already occupied or invalid.2. System displays ' Selected location unavailable '.3. Operator selects another available location.4. Flow resumes at Step 4 of the Main Success Scenario. 2a. Invalid

Field	Specification Barcode/Rfid: 1. System cannot identify the scanned pallet.2. System displays ' Invalid or unregistered pallet. '3. Operator rescans the pallet or reports the issue to the Logistics Supervisor.4. Flow terminates until a valid pallet is identified.
Special Requirements	1. The pallet location lookup and rack-capacity validation should complete within 100 ms under the specified 100,000-bin workload.2. All successful storage and movement records shall contain a timestamp and operator identification.3. Only authorized Warehouse Operators and Logistics Supervisors shall be permitted to perform their respective inventory operations.

4. UML Use-Case Diagram Specification

The UML Use-Case Model illustrates interactions between key actors and the system boundary of the **Warehouse Inventory & Pallet Location Tracker**. It explicitly includes the required «include» and «extend» relationships:

- **Actors:** Warehouse Staff (Primary), Warehouse Manager (Primary), System / Notification Service (Secondary).
- **Primary Use Cases:** Register Inventory, Track Inventory Status, Assign Pallet Location, Monitor Stock Levels & Overdue Movements, Verify & Confirm Inventory Update.
- **«include» Relationship:** UC1 (Register Inventory) «include» UC-Inc (Validate Product & Pallet Details). [Mandatory execution during inventory registration]
- **«extend» Relationship:** UC-Ext (Attach Product Image / Supporting Document) «extend» UC1 (Register Inventory). [Optional execution triggered when the warehouse staff has supporting evidence]
- **«extend» Relationship (Stock Alert):** UC-Ext2 (Trigger Low-Stock Alert) «extend» UC4 (Monitor Stock Levels & Overdue Movements) on condition [Stock Level < Minimum Threshold].

Visual UML Structural Representation

